Multi-Agent Imitation Learning

Foundations of Multi-Agent Imitation Learning: Hardness of non-interactive algorithms and first interactive algorithms.

AUTHOR
Till Freihaut

AFFILIATION
University of Zurich

PUBLISHED
September 9, 2026

READING TIME
12 min

Motivation

The success of Reinforcement Learning (RL) primarily depends on a well-defined reward function. Practitioners spend a significant amount of effort on reward design and shaping, and this is true whether one or many agents are being trained. However, in real systems it is often not true that a reward function is already given or easy to define. Often, we instead have access to expert demonstrations or an expert oracle providing the desired behavior. Imitation Learning (IL) formalizes this by taking the expert as the learning signal and removing the assumption of a pre-defined reward function.

Most recently, IL algorithms gained increasing interest in the context of Large Language Models (LLMs), as supervised fine-tuning and policy distillation methods can be seen from the lens of imitation learning (Agarwal et al., 2024; Rohatgi et al., 2025). Leveraging existing results from IL, and developing new ones, can help in understanding the underlying principles also from a theory point of view.

Broadly, IL splits into two regimes: non-interactive (offline) and interactive (online). The first assumes a fixed dataset of demonstrations, while the second allows the learner to ask the expert for their chosen action at visited states. The most popular non-interactive algorithm is Behavior Cloning (BC) reducing the problem to supervised learning. The classical objection to it is compounding errors (Ross et al., 2011): once the learner reaches states it was not trained on, initially small mistakes accumulate. This is the standard motivation for interactive IL. However, recent work by Foster et al. (2024) establishes that when the learner can accurately represent the expert policy, the apparent gap between BC and interactive IL disappears, making the precise role of interaction less clear than it once seemed. Viano et al. (2026b)

provide further insights that interacting with experts relaxes the representational demands on the learner.

In this blog post, and in our research, we ask the same question in a setting where the answer turns out to be different in kind: Multi-agent Imitation Learning. Many systems of interest involve several agents interacting in a shared environment, such as autonomous driving (Shalev-Shwartz et al., 2016), traffic control (Bakker et al., 2010), and increasingly populations of LLM agents acting alongside one another (Hammond et al., 2025). In these multi-agent systems the concept of what an optimal policy and therefore what an expert is maximizing already changes. Precisely, an expert is no longer simply an agent maximizing an unknown reward, because what counts as good behavior depends on what the other agents do. This necessitates solution concepts from game theory, the most popular being the Nash equilibrium: a strategy from which no agent has an incentive to deviate.

This change in the objective makes learning from expert data significantly harder, and throughout this blog post we explore why.

Setting

Single-agent: MDPs and the value gap

The standard model for sequential decision making is a Markov Decision Process (MDP)

$$\mathcal{M} = (H, \mathcal{S}, \mathcal{A}, P, r, d_0),$$

with horizon H, state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel P, reward function r and initial state distribution d_0. A policy is a collection $\pi = \{\pi_h\}_{h=1}^{H}$ with $\pi_h : \mathcal{S} \to \Delta_{\mathcal{A}}$, and it induces the value function

$$V_h^{\pi}(s) := \mathbb{E}_{\pi}\left[\sum_{t=h}^{H} r_t(S_t, A_t) \,\middle|\, S_h = s\right],$$

and we write $J(\pi) := \langle d_0, V_1^{\pi} \rangle$ for its expected return.

In imitation learning we remove the reward function from the MDP. In the setting we consider there is an unknown expert policy π^E and an unknown MDP $\mathcal{M}$, and the goal is to learn a policy that performs as well as π^E, as measured by expected return. The expert is accessed either as a fixed dataset $\mathcal{D} = \{\tau_k\}_{k=1}^{N}$ of trajectories in the non-interactive setting, or through

queries to π^E at visited states in the interactive setting. Formally, the quantity we control is given by the **value gap**:

$$\text{Val-Gap}(\widehat{\pi}) := J(\pi^E) - J(\widehat{\pi}).$$

Multi-agent: Markov games and the Nash gap

Let us formalize the setting considered in our works. In the following we will consider two agents and a zero-sum setting. Note that all our results can be extended to the n-player general-sum case. The Markov Game is given by

$$\mathcal{G} = (H, \mathcal{S}, \mathcal{A}_1, \mathcal{A}_2, P, r, d_0),$$

where $\mathcal{A}_1$ and $\mathcal{A}_2$ are the action spaces of the two players. Transitions now depend on both actions, $P(s' \mid s, a_1, a_2)$, and the reward is opposing,
$r^1(s, a_1, a_2) = -r^2(s, a_1, a_2) =: r(s, a_1, a_2)$, so that player 1 maximizes what player 2 minimizes. A policy pair (π_1, π_2), with $\pi_{1,h} : \mathcal{S} \to \Delta_{\mathcal{A}_1}$ and $\pi_{2,h} : \mathcal{S} \to \Delta_{\mathcal{A}_2}$ at every stage, induces

$$V_h^{\pi_1,\pi_2}(s) := \mathbb{E}_{\pi_1,\pi_2}\left[\sum_{t=h}^{H} r_t\left(S_t, A_{1,t}, A_{2,t}\right) \,\middle|\, S_h = s\right], \qquad J(\pi_1, \pi_2) := \langle d_0, V_1^{\pi_1,\pi_2}\rangle.$$

We also write $d_h^{\pi_1,\pi_2}(s) := \mathbb{P}_{\pi_1,\pi_2}\left[S_h = s\right]$ for the state occupancy at stage h, which will play a key role in the analysis of non-interactive algorithms.

As the interaction with the environment as well as the value of a learned policy now depends on the policy of the other agent, the idea of optimality changes. However, note that for a fixed policy of one player, the game collapses to an MDP for the other, whose optimal solutions form the best response sets

$$\text{br}(\pi_2) := \arg\max_{\pi_1} J(\pi_1, \pi_2), \qquad \text{br}(\pi_1) := \arg\min_{\pi_2} J(\pi_1, \pi_2).$$

This allows us to define our solution concept and expert assumption considered. A pair (π_1^E, π_2^E) is a **Nash equilibrium** if both policies are best responses to each other. To measure how far an arbitrary pair is from this condition, we use the **Nash gap**

$$\text{Nash-Gap}(\pi_1, \pi_2) := \left\langle d_0, V_1^{\pi_1^\star,\pi_2} - V_1^{\pi_1,\pi_2^\star}\right\rangle, \qquad \pi_1^\star \in \text{br}(\pi_2),\ \pi_2^\star \in \text{br}(\pi_1),$$

which splits into the two players' incentives to deviate,

$$\text{Nash-Gap}(\pi_1, \pi_2) = \underbrace{[J(\pi_1^\star, \pi_2) - J(\pi_1, \pi_2)]}_{\text{what player 1 gains by deviating}} + \underbrace{[J(\pi_1, \pi_2) - J(\pi_1, \pi_2^\star)]}_{\text{what player 2 gains by deviating}}.$$

It satisfies $\text{Nash-Gap}(\pi_1, \pi_2) = 0$ if and only if (π_1, π_2) is a Nash equilibrium, and is strictly positive otherwise. A pair with $\text{Nash-Gap}(\pi_1, \pi_2) \leq \varepsilon$ is called an ε-Nash equilibrium. Multi-agent imitation learning is then the problem of taking data generated by an expert equilibrium or interacting with it (π_1^E, π_2^E) and returning a pair $(\widehat{\pi}_1, \widehat{\pi}_2)$ with small Nash gap.

Comparing the two objectives shows where the difficulty enters. The value gap evaluates the learner's policy as it is. The Nash gap evaluates it against $\text{br}(\widehat{\pi}_2)$ and $\text{br}(\widehat{\pi}_1)$, policies chosen adversarially in response to the learner's own output, and therefore unknown before the learner has produced it.

> **Take away.** Recovering the expert's *value* and recovering the expert's *equilibrium* are different objectives. In the setting of multiple agent, the latter measures the interaction of the agents and is therefore meaningful for the multi-agent setting.

Hardness of non-interactive multi-agent imitation learning

As written earlier, in the single-agent case non-interactive IL can be all you need, and the gain from interactive imitation learning depends on the underlying assumptions (Foster et al., 2024; Viano et al., 2026b). Tang et al. (2024) first identified that once the objective changes to an equilibrium based one, the hardness of the problem increases drastically. In Freihaut et al. (2025, 2026) we generalized these results and characterized exactly what governs the difficulty. To see where it comes from, let us consider a seemingly simple example, a grid world.

Two agents move on a 3×3 grid and race for the goal cell in the top right corner. A state is the *joint* position of both agents, so the game has 72 states, while an expert equilibrium visits only a handful of them. In Figure 1 the expert data covers 2 of the 72 states.

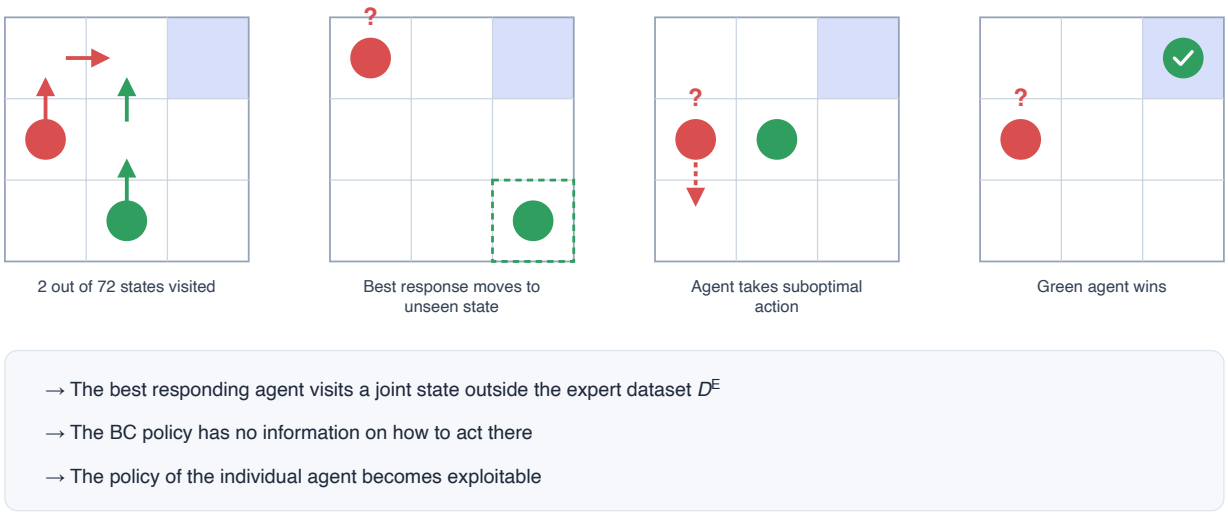


Figure 1. An expert equilibrium visits 2 of 72 joint states. Once the opponent leaves that support, the behavior cloned policy has no information to act on and becomes exploitable.

The intuitive reason why offline IL is hard is that the other player can force you to visit states where no expert behavior has been observed, and where therefore no learning has taken place. Note first that this alone is harmless for the value gap. If the opponent keeps playing the expert policy, the learner is only ever evaluated on the expert's own state distribution, which is precisely the distribution its data came from. The value gap can then be minimized efficiently. The Nash gap is different, because it evaluates the learner against a best response to *its own* output.

More precisely, consider an opponent policy that pushes the learner into an unknown state, even if the required action seems irrational at first glance, for instance moving into a wall and thereby remaining in the same position (Figure 1 center left). Since a state is the joint position of both agents, the resulting state lies outside the support of the expert occupancy measure,

$$s' \notin \bigcup_{h \in [H]} \operatorname{supp}\left(d_h^{\pi_1^E, \pi_2^E}\right) = \{s_1^E, s_2^E\}.$$

On such a state, BC has seen no data at all and falls back to the uniform policy $\widehat{\pi}_1(a_1 \mid s') = 1/|\mathcal{A}_1|$, so one realization is to move down (Figure 1 center right). The best responding agent, meanwhile, takes an action that brings it closer to the goal while keeping the joint state outside the expert data. This can be repeated until the best responding agent reaches the goal and the learned policy is still navigating cluelessly (Figure 1 right).

Formally, the argument above exploits a ratio between two occupancy measures: the one the expert data comes from, and the one the learner is evaluated on once the opponent deviates. Ratios of this kind are commonly referred to as concentrability coefficients, and they measure the overlap between the two distributions. The version that governs the feasibility of non-interactive MAIL lets the opponent best respond to an arbitrary policy, we define as the *all policy deviation* concentrability coefficient,

$$\mathcal{C}_{\max} := \max_{h \in [H]} \max_{\pi_1, \pi_2} \max \left\{ \max_{\pi_2^\star \in \mathrm{br}(\pi_1)} \left\| \frac{d_h^{\pi_1^E, \pi_2^\star}}{d_h^{\pi_1^E, \pi_2^E}} \right\|_\infty, \max_{\pi_1^\star \in \mathrm{br}(\pi_2)} \left\| \frac{d_h^{\pi_1^\star, \pi_2^E}}{d_h^{\pi_1^E, \pi_2^E}} \right\|_\infty \right\}.$$

Intuitively, this coefficient describes exactly the overlap of observed expert states visited and the states a best response to an arbitrary policy can force us into. It becomes infinite precisely when some state that is reachable under a deviation has zero probability under the expert equilibrium, which is what happens in Figure 1. Note that the best responses inside the maximum are taken against an *arbitrary* policy and not against the expert. This is what makes $\mathcal{C}_{\max}$ also account for deviations that look irrational at first glance, such as deliberately walking into a wall in order to leave the expert's support.

In Freihaut et al. (2025, 2026) we turned this intuition into lower bounds. We construct zero-sum Markov games, one of them with three states only, in which $\mathcal{C}_{\max}$ is infinite and every non-interactive imitation learning algorithm suffers a Nash gap of order of the horizon H. This is not an artifact of the learner knowing too little about the game: the result holds even with infinite expert data, even if the learner knows the transition dynamics, and even if it matches the expert occupancy measure perfectly. When $\mathcal{C}_{\max}$ is finite, the same construction gives a statistical lower bound of $\Omega\left(\mathcal{C}_{\max}\varepsilon^{-2}\right)$ on the number of expert trajectories needed by *any* non-interactive MAIL algorithm. Since BC attains $\widetilde{\mathcal{O}}\left(\mathcal{C}_{\max}^2\varepsilon^{-2}\right)$, it is rate-optimal in ε: no non-interactive algorithm can improve the dependence on the accuracy, and the remaining gap is only polynomial in $\mathcal{C}_{\max}$ and in the parameters of the game.

The role of $\mathcal{C}_{\max}$ is visible directly in the behavior of BC. Figure 2 runs behavior cloning on a family of lower bound Markov games that differ only in their concentrability coefficient, and reports the Nash gap of the learned pair as the number of expert trajectories grows.

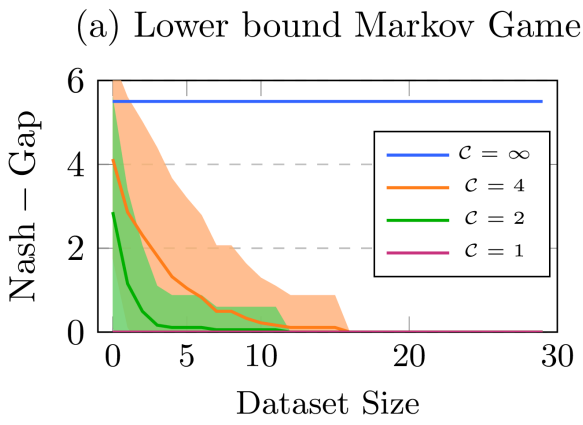


Figure 2. Behavior cloning on lower bound Markov games with concentrability coefficient $\mathcal{C}_{\max} \in \{1, 2, 4, \infty\}$. Shaded regions show the spread across seeds. The finite-$\mathcal{C}_{\max}$ curves reach zero Nash gap, and the more expert data they need the larger $\mathcal{C}_{\max}$ is; at $\mathcal{C}_{\max} = \infty$ the curve never moves.

The picture matches the theory line for line. Whenever $\mathcal{C}_{\max}$ is finite the Nash gap does go to zero, so offline data is not useless, but the coefficient sets the price: $\mathcal{C}_{\max} = 1$ is solved immediately, $\mathcal{C}_{\max} = 2$ needs a handful of trajectories and $\mathcal{C}_{\max} = 4$ roughly three times as many. The blue curve is the case the lower bound is about. With $\mathcal{C}_{\max} = \infty$ the Nash gap sits flat at its initial value across the whole range, and no amount of additional expert data moves it, because the states on which the learner is exploitable are ones the expert equilibrium simply never visits.

> **Take away.** In non-interactive MAIL, the performance of the algorithms is driven by $\mathcal{C}_{\max}$ and in case it is infinite, all non-interactive algorithms fail to learn.

Finally, it is worth emphasizing that this failure is not confined to artificial lower bound constructions. In Viano et al. (2026a) we observe the same effect in classical games such as Tic-Tac-Toe and Connect Four, where interactive MAIL algorithms clearly outperform behavior cloning.

Can better representations rescue offline MAIL?

Looking again at Figure 1, one might object that the situation is not nearly as hopeless as the theory suggests. A human player dropped into the deviated state would not be lost. Their internal representation of the game is not the raw joint position of both agents, it is something closer to *how far am I from the goal and where is my opponent relative to me*, and under that

representation the unseen state looks very much like states that were observed. The expert data should then be informative about it, even though the state itself never occurred.

This intuition can be made precise. Representations of this kind are formalized by *linear Markov games* (Jin et al., 2019), in which transitions and rewards, averaged over the expert policy of the other player, are linear in a known feature map φ. Tabular Markov games are the special case of one-hot features, so this is a strict generalization of the setting above. In Viano et al. (2026a) we show that under this structure the concentrability coefficient can be stated at the level of features rather than states. For a single player, and suppressing the outer maximum over the two players, it takes the form

$$\mathcal{C}_{\varphi,\max} := \max_{h\in[H]} \max_{\pi_2} \max_{\pi_1^\star \in \mathrm{br}(\pi_2)} \left\| \varphi_h^{\pi_1^\star,\pi_2^E} \right\|_{\Lambda_{E,h}^{-1}}, \qquad \Lambda_{E,h} := \mathbb{E}_{\pi_1^E,\pi_2^E}\left[\varphi\left(S_h, A_{1,h}\right)\varphi(S_h,A_{1,h})^\top\right] + \lambda I,$$

where $\varphi_h^{\pi_1^\star,\pi_2^E}$ is the expected feature vector generated by the deviating pair at stage h. It is the same overlap measure as before, only now the deviation has to be poorly covered in *feature space* rather than in state space. Since two states with non-orthogonal features carry information about each other, the learner can act sensibly in states it has never visited, and $\mathcal{C}_{\varphi,\max} \leq \mathcal{C}_{\max}$, potentially by a wide margin. With one-hot features the two coincide, which is exactly why the tabular case looked so bad. Behavior cloning then needs $\widetilde{\mathcal{O}}\left(\mathcal{C}_{\varphi,\max}^2 d\,\varepsilon^{-2}\right)$ expert trajectories, with the feature dimension d in place of the size of the state space.

So the answer is yes, representations help. But they are not a free lunch, and the reason is a tension that is easy to miss. Consider the most similar features imaginable, $\varphi(s,a_1) = c$ for every state and action. Every state then looks identical to every other, nothing is ever out of distribution, and $\mathcal{C}_{\varphi,\max} = 1$ attains its smallest possible value. These features are of course useless: they are not expressive enough to satisfy the linear Markov game assumption or to realize the expert policy. Good features have to do two things at once, generalize widely enough to keep $\mathcal{C}_{\varphi,\max}$ small, and remain expressive enough to represent the game and the expert. Coming up with them is an environment specific design problem, and in general a hard one.

We tested these theoretical findings for different feature maps in a practical setup. Figure 3 reports behavior cloning in a bigger grid world environment under three feature maps: tabular one-hot features with $d = 288$, relational features built from relative state similarities with $d = 80$, and a linear neural network with $d = 262$.

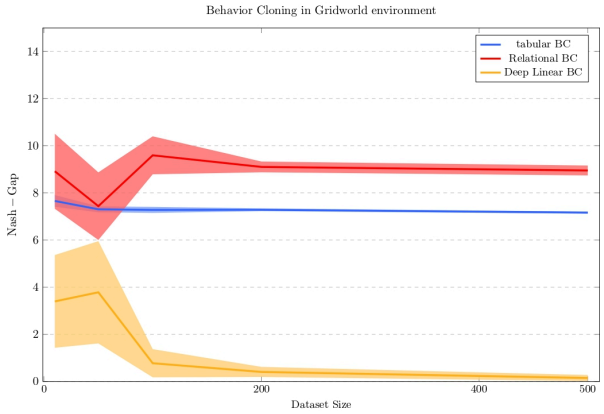


Figure 3. Behavior cloning in the 3×3 gridworld. Tabular and relational features leave the Nash gap flat in the dataset size, while learned linear features drive it to zero.

Tabular BC does not improve at all as the dataset grows, its Nash gap sitting flat at roughly 7, which is the empirical signature of an unbounded $\mathcal{C}_{\max}$. More data does not help, exactly as the lower bound predicts. The relational features are close in spirit to the relative representation we appealed to above, and they are *worse* than tabular, flat at around 9. Only the learned linear features drive the Nash gap towards zero. In all three cases BC fits its training data perfectly, so what separates the curves is not optimization but $\mathcal{C}_{\varphi,\max}$.

> **Take away.** Better representations change the constant, not the structure. A feature level concentrability coefficient can be dramatically smaller than its tabular counterpart, and Figure 3 shows that the choice of features decides whether non-interactive algorithms learn anything at all. But it remains a coverage assumption, it can still be infinite, and no choice of features makes an offline algorithm concentrability free.

Minimizing the Nash gap with interactive experts

In the previous sections we have learned that non-interactive imitation learning faces significant challenges, as the performance of every non-interactive algorithm depends on $\mathcal{C}_{\max}$, or on its relaxation to feature space. As this would be a sad end to the story, we are happy to provide effective algorithms in the interactive MAIL regime.

The overall intuition behind these algorithms is to split learning into two phases, an exploration phase and an imitation phase. The goal of the first phase is to explore the environment in such a way that the states in which the agent is potentially highly exploitable

are already covered, so that in the second phase the expert policies can be learned in those states.

For this, let us revisit the environments used for the lower bounds. The underlying problem was that if there exists a state which has not been visited, or whose representation is orthogonal to that of the states we did visit, then the learner does not know how to act and the opponent can exploit the learned behavior. We would therefore like to visit these states and ask the expert how to behave in them, removing the exploitable part of the policy. The difficulty is that we do not know the reward function of either player, and so we do not know a priori which states we should query the expert in. The main idea is to leverage techniques from **reward-free** reinforcement learning (Jin et al., 2020; Wang et al., 2020; Kaufmann et al., 2021). At a high level, the goal there is to explore all important states, meaning all states reachable with a certain probability, and to guarantee that once *any* reward function is revealed, a near-optimal policy can be computed from the collected data. Since the existing reward-free results are limited to the single-agent setting, we need to adapt them, and we do so by divide and conquer.

More precisely, note that in a two player game, fixing the policy of one agent induces a single-agent environment with an altered transition kernel. Since we are allowed to query the experts, we can fix the expert policy π_2^E of player 2 and obtain the *expert induced MDP* for player 1, whose transitions are given by

$$P^{\pi_2^E}\left(s' \mid s, a_1\right) := \sum_{a_2 \in \mathcal{A}_2} \pi_2^E\left(a_2 \mid s\right) P\left(s' \mid s, a_1, a_2\right).$$

Each player therefore has its own expert induced MDP, and within it the problem is a single-agent one, so the classical reward-free machinery applies. Concretely, we follow the reward-free exploration scheme of Jin et al. (2020) to cover all important states of each expert induced MDP. Instead of being handed an arbitrary reward function afterwards, however, we use the exploration phase for a different purpose: while one agent explores, the expert on the other side is being queried at every visited state, so the exploration phase silently produces two expert datasets $\mathcal{D}^{\pi_1^E}$ and $\mathcal{D}^{\pi_2^E}$. In contrast to the expert versus expert data used by non-interactive algorithms, these datasets consist of expert versus explorer trajectories and are therefore far more diverse. It then remains to split the Nash gap into the two subproblems and to *conquer* each of them with plain behavior cloning on the corresponding dataset.

That decomposition follows from the definition of the Nash equilibrium, the performance difference lemma (Kakade and Langford, 2002), Hölder's inequality and the worst case bound

on the value functions. Writing $\mathrm{TV}(\pi, \pi')(s) := \sum_{a \in \mathcal{A}_n} |\pi(a \mid s) - \pi'(a \mid s)|$ for two policies of the same player n, it reads

$$\left\langle d_0,\, V_1^{\pi_1^\star, \widehat{\pi}_2} - V_1^{\widehat{\pi}_1, \pi_2^\star} \right\rangle \;\le\; 2H \sum_{h=1}^{H} \sum_{\pi \in \{\widehat{\pi}_1, \widehat{\pi}_2\}} \mathrm{Err}_h(\pi),$$

where the two error terms are

$$\mathrm{Err}_h\left(\widehat{\pi}_2\right) := \max_{\pi_1 \in \mathrm{br}(\widehat{\pi}_2)} \mathbb{E}_{s \sim d_h^{\pi_1, \pi_2^E}} \left[\mathrm{TV}\left(\pi_{2,h}^E, \widehat{\pi}_{2,h}\right)(s)\right], \qquad \mathrm{Err}_h\left(\widehat{\pi}_1\right) := \max_{\pi_2 \in \mathrm{br}(\widehat{\pi}_1)} \mathbb{E}_{s \sim d_h^{\pi_1^E, \pi_2}} \left[\mathrm{TV}\left(\pi_{1,h}^E, \widehat{\pi}_{1,h}\right)(s)\right].$$

The left hand side is exactly the Nash gap of the output pair. What makes this decomposition the right one is where the expectations are taken. Each error term is measured under an occupancy in which *one player follows the expert while the other is arbitrary*, and that is precisely the distribution generated by exploring the corresponding expert induced MDP. The datasets built in phase one are thus supported on the very states on which phase two is evaluated, which is why no concentrability coefficient appears.

This is the algorithm we call MAIL-WARM. It returns an ε-Nash equilibrium using $\widetilde{\mathcal{O}}\left(\varepsilon^{-2}\right)$ expert queries, matching the lower bound, so the dependence on the accuracy can no longer be improved even with interaction.

Note also that exploration is carried out in the two induced MDPs rather than in the game itself, which is what keeps the algorithm free of the curse of multi-agents (Wang et al., 2023; Cui et al., 2023), meaning that even for n-player general-sum games the sample complexity does not scale exponentially in the number of players. In that setting the expert induced MDP for player i is obtained by fixing every other expert π_{-i}^E, the Nash gap becomes the largest gain any single player can obtain by unilaterally deviating, and the same two-phase scheme returns an ε-Nash equilibrium from

$$N = \mathcal{O}\left(\frac{n^4 H^6 S^3 A_{\max}^3 \log\left(S/\delta_{\mathrm{fail}}\right)}{\varepsilon^2}\right)$$

trajectories, and therefore $\mathcal{O}(NH)$ expert queries, where $S := |\mathcal{S}|$ and $A_{\max} := \max_{i \in [n]} |\mathcal{A}_i|$. Two things are worth reading off this bound. The dependence on the number of players is polynomial rather than exponential, and $A_{\max}$ is the largest *individual* action space, not the joint one, whose cardinality $\prod_i |\mathcal{A}_i|$ is itself exponential in n.

Scaling beyond the tabular case

The warm-up phase described above loops over state and stage pairs: for every (s, h) it solves a reward maximization problem whose reward is one for reaching s at step h. In a tabular game this is fine, but it instantiates SH separate RL problems and its guarantees scale with S, so it is hopeless once the state space is large.

In Viano et al. (2026a) we replace this with a single reward-free run per player inside the induced *linear* MDP. Rather than targeting one state at a time, the explorer is rewarded for reaching feature directions that the data collected so far covers poorly, that is, at episode k it maximizes the bonus $\|\varphi(s,a)\|_{(\Lambda^k)^{-1}}$. We call the resulting procedure LSVI-UCB-ZERO, since it amounts to running LSVI-UCB (Jin et al., 2019) everywhere with zero reward and a slightly enlarged bonus, and the imitation phase is again plain behavior cloning on the dataset it produces. Running it for

$$K = \widetilde{\mathcal{O}}\left(\frac{H^6 d^4 B^4 \log\left(A_{\max} B_\theta \delta^{-1}\right)}{\varepsilon^2}\right)$$

episodes, and therefore with KH expert queries, returns an ε-Nash equilibrium with probability at least $1-\delta$, where B and B_θ are the problem dependent bounds on the linear parameters.

> **Take away.** In the interactive setting the concentrability coefficient disappears entirely, in the linear case as well as in the tabular one. What is more, the sample complexity now scales with the feature dimension d and no longer with the number of states of the game.

A note on the horizon

We have presented everything in the finite-horizon setting. The hardness results of the previous sections were originally derived in the discounted infinite-horizon setting, with the effective horizon $(1-\gamma)^{-1}$ playing the role of H, and they transfer to the finite-horizon formulation used here.

The reverse direction is less immediate. The warm-up phase of MAIL-WARM inherits regret guarantees from a finite-horizon reward-free subroutine, namely EULER (Zanette and Brunskill, 2019), which cannot be easily extened to the discounted setting. Therefore, it was left open in Freihaut et al. (2026). It is answered affirmatively by the linear

algorithm of Viano et al. (2026a): the same two-phase design goes through once the exploration subroutine is replaced by one with optimal regret in discounted linear MDPs (Moulin et al., 2025), under a mild normalization of the feature map. Since tabular Markov games are linear Markov games, this also settles the discounted case we started from.

Beyond linear: a step towards deep multi-agent imitation learning

Our theoretical analysis covers tabular and linear Markov games, and the natural next step is to carry the algorithmic ideas over to a deep MAIL algorithm. We take that step in Viano et al. (2026a) with DQN-Explore-BC.

The one component that does not survive the move to general function approximation is the exploration bonus, which in the linear case is built from the given feature map. We replace those features with their function approximation analogue, the last layer of a neural network trained to predict the cumulative future exploratory reward, that is, a critic network in deep RL jargon. This introduces a difficulty that is absent in the linear case: the features are no longer fixed, they change as the network is updated, so the bonus has to be recomputed on the historical data every time the feature map moves.

We evaluate the algorithm on two classical games, Tic-Tac-Toe and Connect Four. Both are strongly solved, so we can use optimal solvers as expert queries. In Tic-Tac-Toe, where optimal play ends in a draw, the exploitability of the learned policy can be computed exactly.

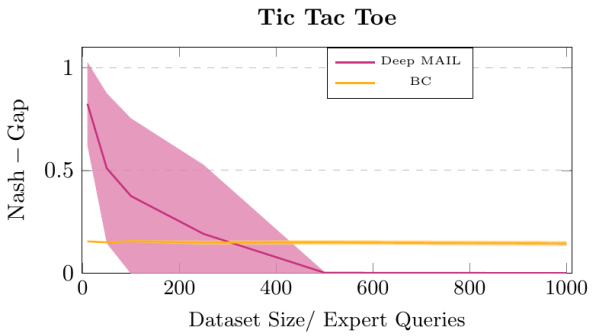


Figure 4. Tic-Tac-Toe. Behavior cloning plateaus at a strictly positive Nash gap, while deep MAIL drives the exploitability of its policy to zero.

The two curves separate in the way the theory predicts. Behavior cloning sits flat at a Nash gap and does not move as the dataset grows, which is the offline barrier of the earlier

sections appearing in a real game rather than in a lower bound construction. Deep MAIL starts out considerably worse, at around 0.8, since its early queries are spent exploring rather than imitating, overtakes BC after roughly 300 queries, and reaches zero exploitability at around 500. Spending part of the query budget on exploration is what makes the difference between a policy that stops improving and one that becomes unexploitable.

In Connect Four the state space is far too large to compute exploitability exactly, so we instead measure the win rate of the learned first player against opponents of decreasing strength, from an approximate best response to solvers perturbed with increasing amounts of noise.

OPPONENT	DQN-EXPLORE-BC	BC
Approximate best response	**0.21 ± 0.04**	0.00 ± 0.00
Solver, noise 1	**0.32 ± 0.04**	0.00 ± 0.00
Solver, noise 2	**0.60 ± 0.04**	0.05 ± 0.01
Solver, noise 3	**0.81 ± 0.01**	0.21 ± 0.02
Solver, noise 4	**0.92 ± 0.00**	0.40 ± 0.04
Solver, noise 5	**0.97 ± 0.01**	0.54 ± 0.05

Table 1. Connect Four win rates against opponents of varying optimality. DQN-Explore-BC outperforms BC at every level.

Summary

In our recent works, we have tackled the problem of multi-agent imitation learning and contributed to the theoretical understanding of this setting and made the first step to deep multi-agent imitation learning. This opens up many interesting directions of future work especially in applied settings which we rarely explored yet. If you have questions or want to collaborate, feel free to reach out to any of the authors!

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